

IMO Training – Geometry

More Geometry via Complex Numbers

Longtin Chan
longtinclt@gmail.com

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Notes

Unless otherwise specified, let O be the origin, and let the smaller case letter denote the corresponding complex number on the complex plane of the vertex. For example, the complex number of point A is denoted as a .

Definition. (Straight Line) The straight line AB is denoted by $\frac{z-a}{b-a} = \frac{\bar{z}-\bar{a}}{\bar{b}-\bar{a}}$.

Definition. (Unit Circle) The unit circle is denoted by $|z| = 1$.

Proposition. (Rotation) If C is the image of rotating B with respect to A by angle θ , then $c = a + (b-a)e^{i\theta}$.

Proposition. (Chord) If AB is a chord on the unit circle, then any point Z on AB satisfies

$$z + ab\bar{z} = a + b.$$

Proposition. (Tangent) If A is a point on the unit circle, then any point Z on the tangent at A satisfies

$$z + a^2\bar{z} = 2a.$$

Furthermore, the intersecting point P of tangents at A and B is $p = \frac{2ab}{a+b}$.

Proposition. If A, B, C are on the unit circle, then the orthocentre is $h = a + b + c$.

Proposition. If A, B, C are on the unit circle, then for incentre I of $\triangle ABC$ such that AI, BI, CI meets the unit circle again at D, E, F , there exists $x, y, z \in \mathbb{C}$ such that

$$a = x^2, b = y^2, c = z^2, d = -yz, e = -zx, f = -xy,$$

and I is given by $-(xy + yz + zx)$. (I is the orthocentre of $\triangle DEF$)

Proposition. The circumcentre of $\triangle ABC$ is

$$\begin{vmatrix} a & a\bar{a} & 1 \\ b & b\bar{b} & 1 \\ c & c\bar{c} & 1 \end{vmatrix} \div \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}.$$

Examples

1. (Simson Line) Let P be a point on the circumcircle of $\triangle ABC$. Let D, E, F be the feet of perpendiculars from P to lines BC, CA, AB respectively. Prove that D, E, F are collinear. Furthermore, prove that line DEF bisects PH , where H is the orthocentre of $\triangle ABC$.
2. (Nine-point circle) Let H, O be the orthocentre and the circumcentre of $\triangle ABC$ respectively. Let D, E, F be the midpoints of BC, CA, AB and let A', B', C' be the feet of altitudes from A, B, C respectively. Let P, Q, S be the midpoints of AH, BH, CH respectively. Prove that $D, E, F, A', B', C', P, Q, S$ lie on the same circle. Also, prove that the centre of this circle is the midpoint of HO , and the radius is half of the circumradius of $\triangle ABC$.
3. (RMM 2019 Problem 2) Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC . Denote by ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D . Prove that PE is tangent to Ω .
4. (Taiwan TST 2014) Let ABC be a triangle with incenter I , and suppose the incircle is tangent to CA and AB at E and F . Denote by G and H the reflections of E and F over I . Let Q be the intersection of BC with GH , and let M be the midpoint of BC . Prove that IQ and IM are perpendicular.
5. (APMO 2013) Let $ABCD$ be a quadrilateral inscribed in a circle ω , and let P be a point on the extension of AC such that PB and PD are tangent to ω . The tangent at C intersects PD at Q and the line AD at R . Let E be the second point of intersection between AQ and ω . Prove that B, E, R are collinear.
6. (IMOSL 2016 G5) Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC . A circle ω with centre S passes through A and D , and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC , and let M be the midpoint of BC . Prove that the circumcentre of triangle $XS Y$ is equidistant from P and M .
7. (IMOSL 2017 G3) Let O be the circumcenter of an acute triangle ABC . Line OA intersects the altitudes of ABC through B and C at P and Q , respectively. The altitudes meet at H . Prove that the circumcenter of triangle PQH lies on a median of triangle ABC .
8. (IMOSL 2018 G2) Let ABC be a triangle with $AB = AC$, and let M be the midpoint of BC . Let P be a point such that $PB < PC$ and PA is parallel to BC . Let X and Y be points on the lines PB and PC , respectively, so that B lies on the segment PX , C lies on the segment PY , and $\angle PXM = \angle PYM$. Prove that the quadrilateral $APXY$ is cyclic.
9. (IMOSL 2021 G4) Let $ABCD$ be a quadrilateral inscribed in a circle Ω . Let the tangent to Ω at D meet rays BA and BC at E and F , respectively. A point T is chosen inside $\triangle ABC$ so that $TE \parallel CD$ and $TF \parallel AD$. Let $K \neq D$ be a point on segment DF satisfying $TD = TK$. Prove that lines AC, DT , and BK are concurrent.

10. (IMO 2010 Problem 2) Given a triangle ABC , with I as its incenter and Γ as its circumcircle, AI intersects Γ again at D . Let E be a point on the arc BDC , and F a point on the segment BC , such that $\angle BAF = \angle CAE < \frac{1}{2}\angle BAC$. If G is the midpoint of IF , prove that the meeting point of the lines EI and DG lies on Γ .
11. (IMO 2015 Problem 3) Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A, B, C, K and Q are all different and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other.
12. (IMO 2019 Problem 6) Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA , and AB at D, E , and F , respectively. The line through D perpendicular to EF meets ω at R . Line AR meets ω again at P . The circumcircles of triangle PCE and PBF meet again at Q . Prove that lines DI and PQ meet on the line through A perpendicular to AI .